

MSDS 498 CAPSTONE

Client Check-In: Week 5

The data dictionary is ready for your review — plus three decisions we need from you

Where the build stands

One frozen, versioned dataset — everything downstream runs off it

34,392

high schools in the frozen
modeling dataset (v1)

57

variables, every one
documented

61

tables in the database,
rebuilt end to end

72 / 0

tests passing / skipped

The pipeline, end to end

1

Raw

one table per source — NCES, Census,
NAEP, ISBE, IB, CPS, College Board, CRDC

2

Clean

typed casts, suppression codes nulled

3

Combined

school-level backbone + CEEB crosswalks

4

Modeled

features → rigor tiers → clustering,
now queryable in the database

Don't want to run any of it? Every table is also exported to CSV — open the files directly.

Shipped since we last met

Three bug fixes, one gap closed, and the whole pipeline verified from scratch

1

Fixed a CEEB fan-out bug

Match rate corrected 73% → 64.5%. The old 73% included ~2,072 false matches. We reported the higher number to you before we found it.

2

Fixed the district-ID and sector bugs

The first was blocking per-student funding (Goal 4) entirely; the second was silently hiding IB flags.

3

Made the modeling dataset queryable

It previously existed only as a file — the database stopped one step short of it. Now loaded as database tables alongside everything else.

4

Verified the rebuild end to end

Full pipeline run from raw sources: all six stages complete, 61 tables, every test passing with none skipped.

"A data dictionary of all their variables — do we have this?"

Yes. It's ready for you and Adam now.

The raw joined table

127

variables

The modeling feature set

57

variables — every column covered

The dataset itself

34,392

rows, frozen and versioned

Every variable carries: source · grain · vintage · confidence · range · % populated · description

The ask

Can you and Adam read through these and send questions back? It's the fastest way to catch a variable we've mislabeled before it propagates into the modeling.

One number, three denominators

You may see two different match rates in our materials — here's why both are right

The 34,392 rows, decomposed

Org-only rows (no school record) <i>can never match, by construction</i>	12,560
School-side record present	21,832
→ matched to your org data	16,111
→ unmatched	5,721

64.5%

the crosswalk's own performance

Use for the join

16,508 / 25,577 — of schools we started with, how many matched

73.8%

within the modeling dataset

Use for the dataset

16,111 / 21,832 — of rows that could match, how many did

46.8%

share of all modeling rows

Don't quote alone

16,111 / 34,392 — penalizes the join for rows it was never offered

The numerator is identical in both figures.

16,508 matched schools carry all the way through; the freeze then drops 397 of them for being too small or not high schools, leaving 16,111. Nothing about the matching changed — only what we divided by.

Full derivation in docs/MATCH_RATE_RECONCILIATION.md — flagging it here so the two numbers don't look like an error if you see them side by side.

Different sources, different update cycles

Your concern from last week — it's really two questions, and they have different answers

1 When we pulled the file

- Solved — answerable from our own repository history, no one else needed.
- 22 of 23 source files now carry a real first- and last-updated date.
- Regenerates automatically whenever a source is refreshed.

2 When the data was measured

- Still open — this is the one we can't answer ourselves.
- For your 56 fields we can confirm the export date (June 24) but not when each statistic was actually measured.
- This is exactly the "old section vs. new section" problem — we built the column to hold the answer, we just don't have the answer.

13 of 57 variables are waiting on a measurement date — 11 of those need Sheng, not you. It's a short, specific list now rather than an open-ended question.

Three questions we closed without you

All three were on last week's list for Bob — none of them still needs your time

1

Historical rigor labels

Not needed after all. The rigor tier is a transparent weighted composite, not a model trained on past ground truth — so there's nothing to label.

2

Are the socio-context fields need-coded?

Confirmed from the data itself. All five share a 1–100 scale and move together (correlations 0.38–0.91); two check out independently against Census figures. So all five are coded the same way.

3

The custom ID in your org export

Investigated directly. It's 76.5% populated, matches no federal identifier we hold, and has 84 duplicate values — an internal ID from whatever system produced the export, with no cross-referencing value here.

Our default going forward: if a question is answerable from the data, we answer it and tell you what we found — and only escalate what genuinely needs your knowledge of your systems.

One thing we found in your export

Surfaced by accident while investigating the custom ID — flagged, not corrected

The pattern

Two clearly different schools whose CEEB codes are related by an exact leading-zero / trailing-zero shift:

050003 **vs.** 500030

The classic signature of a 6-digit zero-padded code stored as a number somewhere upstream, then re-padded on the wrong side.

62

confirmed pairs

duplicate-ID pairs fitting the pattern exactly

644

flagged across the file

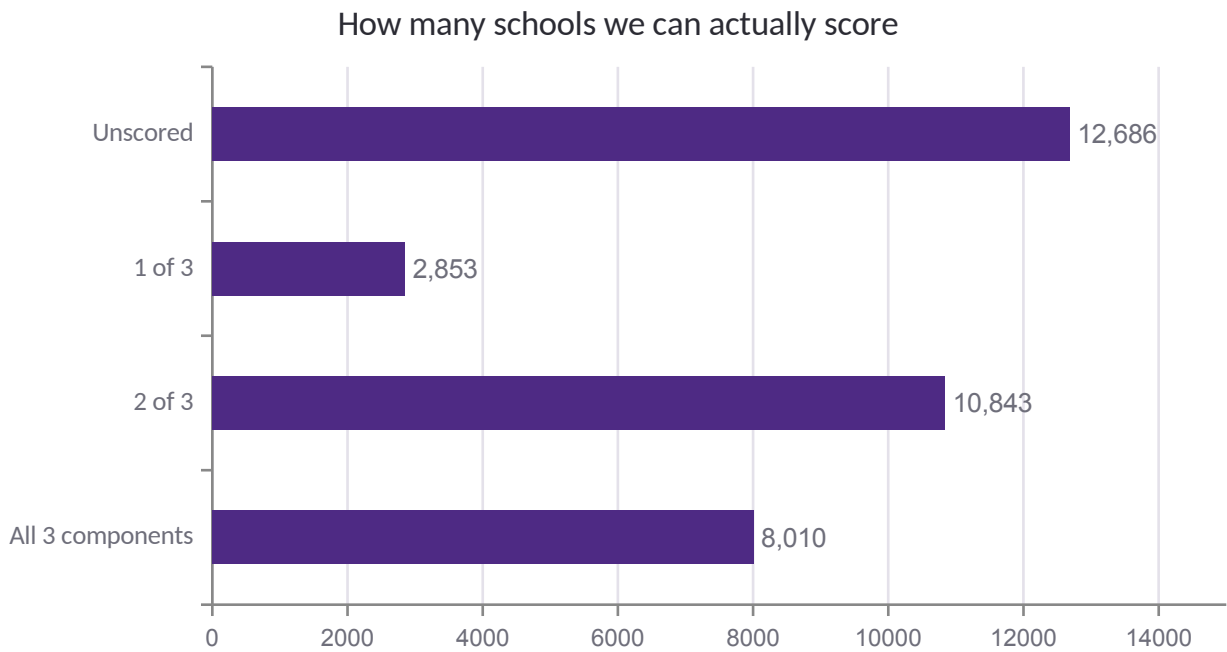
codes with a same-file counterpart fitting the pattern — some fraction of those will be coincidental

Flagged in the data, not auto-corrected — there's no reliable way to tell which side of a pair is the right one without knowing your source system.

Ask for Bob: do you know which export step could introduce this? If it's fixable upstream that's cleaner than us guessing downstream — and it may affect your production data too, not just our copy.

Rigor classification — first pass

A transparent weighted composite across AP, advanced coursework, and test participation



Schools with none of the three components are left unscored — not quietly defaulted to a middle tier.

The number worth your attention

52.5%

of covered schools would land in a different rigor tier if federal civil-rights data became unavailable and we fell back to your data alone.

And one reassuring one

The tier isn't simply re-deriving family income: its correlation with child poverty is -0.07 and with school funding 0.03 – 0.08 . Weak, and in the directions you'd expect.

What we need from you

Ordered by how much each is holding us up

1

Measurement dates for your 56 fields

When was each statistic actually measured — average SAT, AP participation, and so on? We have the export date, not the measurement dates.

Blocking

2

Sign-off on the funding estimate

Is district revenue divided by school-age population an acceptable national stand-in for per-pupil spending? Illinois has a true figure; the national one is an estimate.

Decision

3

A tie-breaker rule for AP data

When the federal AP figures and yours disagree, which one wins — or do we carry both?

Decision

4

College Board institutional access

Still on your side. Until it lands, AP data stays national and state-level only — nothing school-by-school.

Carried over

Separately: 11 more measurement-date questions are for Sheng, and the federal ID re-pull is ours — noted so neither looks like a gap on your side.

Next two weeks

What we do regardless, and what's waiting on answers

Moving regardless

- Re-pull the federal school file with full-length IDs — unblocks district funding for every public school
- Second pass on the rigor tier cut-points (currently even fifths — transparent, but an arbitrary choice)
- Widen clustering coverage beyond the 17% of schools with complete data on all features
- Decide how to report the schools we can't score at all, rather than dropping them silently

Waiting on you

- Measurement dates — so the dictionary stops reading "not confirmed"
- Funding sign-off — Goal 4 either ships nationally or stays Illinois-only
- AP tie-breaker — determines what feeds the rigor score
- Dictionary review from you and Adam — catches anything mislabeled before it propagates

Working principle: sources disagree and get more useful over time — so every merge carries a match rate, a source, and an as-of date rather than one silent answer.